



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Angel Tripana
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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Summary of methodologies**

- Data Collection through API and Web scraping
- Data cleaning with SQL and Python
- Data analysis with Python
- Data visualization with Python
- Machine Learning Prediction

- **Summary of all results**

- Collect data from public sources
- Exploratory Data Analysis result
- Interactive analytics dashboards
- Predictive ML analytic models

Introduction

- **Project background and context**

SpaceY company analyze SpaceX launches results for Falcon 9 rockets in first stage landing to be an alternative.

- **Problems you want to find answers**

Determine the launch costs based on the reuse of rockets in the first stage and find best sites and conditions to makes launches.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Data was obtained from two public sources
 - SpaceX API (<https://api.spacexdata.com/v4/rockets/>)
 - Web scraping (https://en.wikipedia.org/wiki/List_of_Falcon/_9/_and_Falcon_Heavy_launches)
- Perform data wrangling

Landing outcome label based on outcome data after summarizing and analyzing features.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models

Normalized and split data on train and test sets, evaluated data on different classification models to find the best performance.

Data Collection

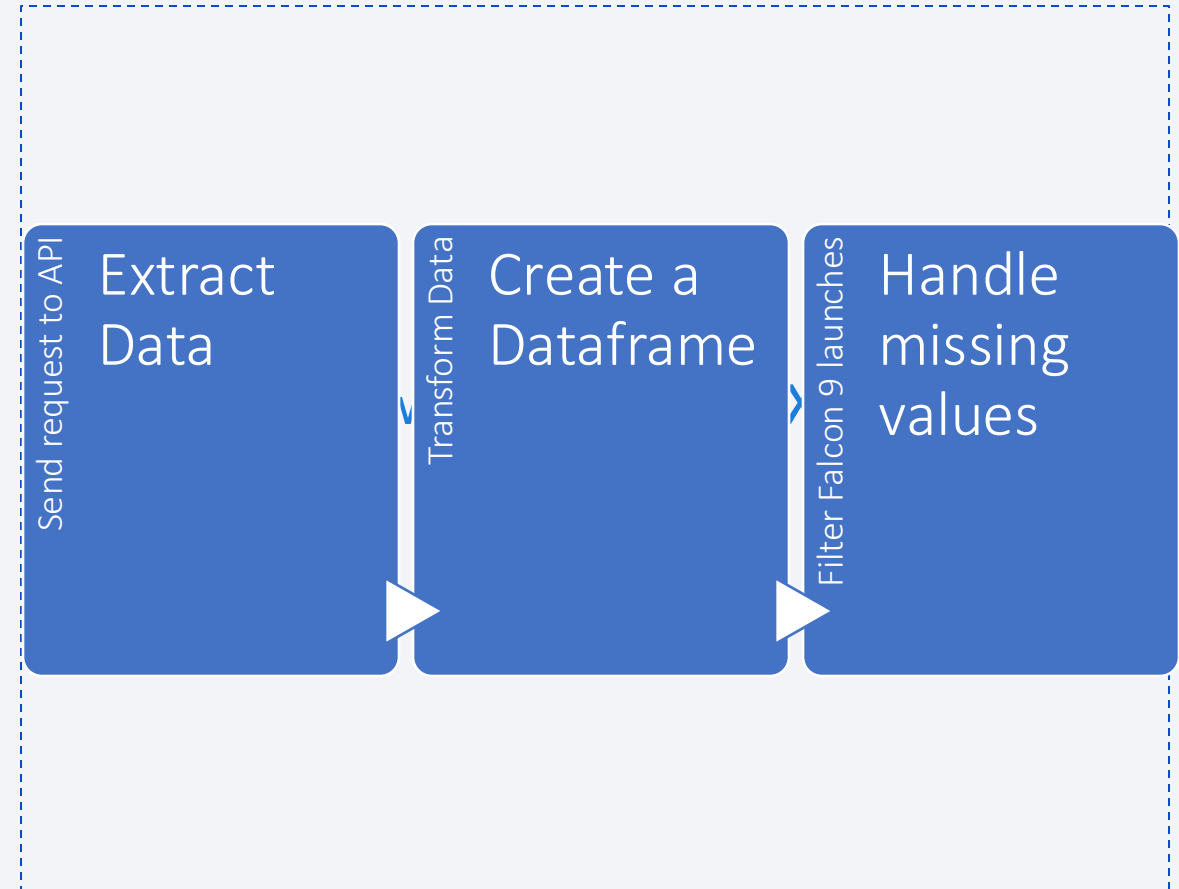
- Describe how data sets were collected.

Data was collected from two public data sources.

- SpaceX API (Using public API)
- Wikipedia (Web scraping)

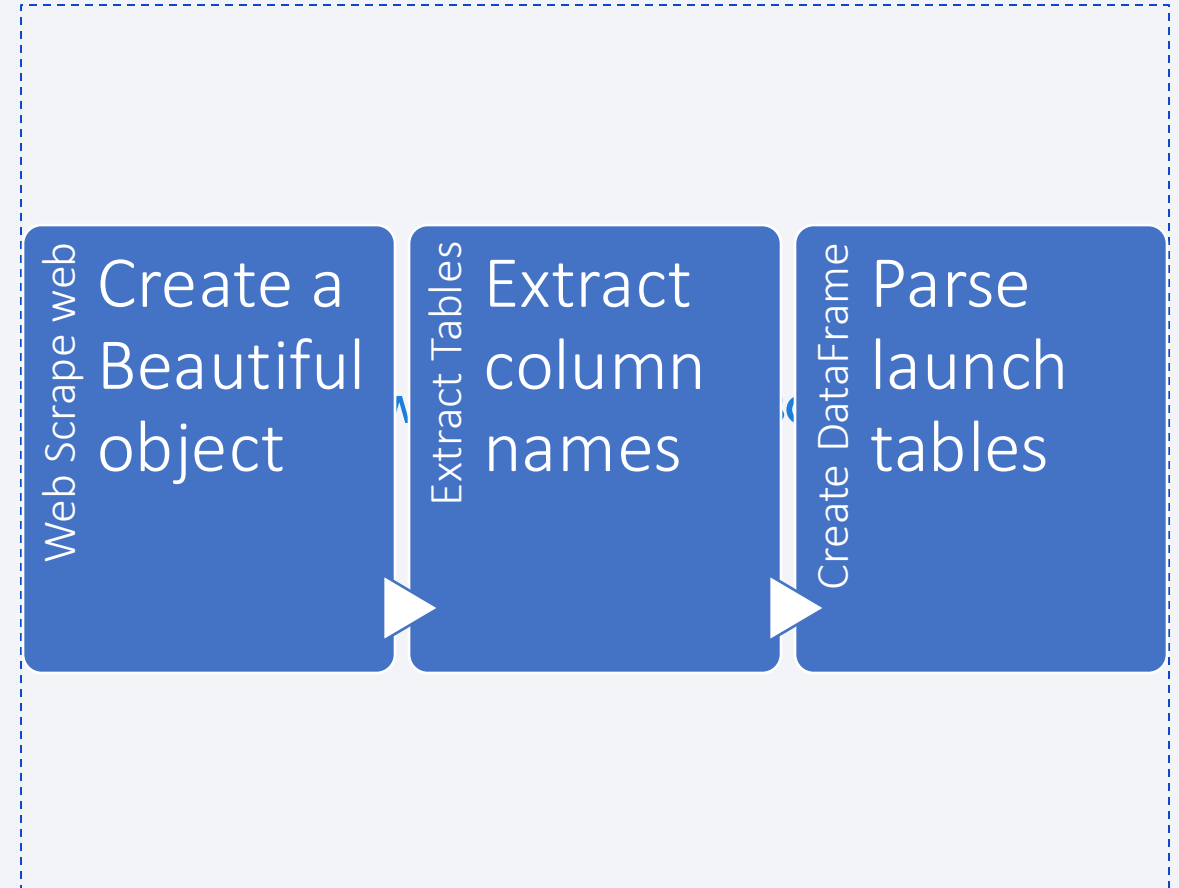
Data Collection – SpaceX API

- Public API from SpaceX
- [GitHub Link](#)



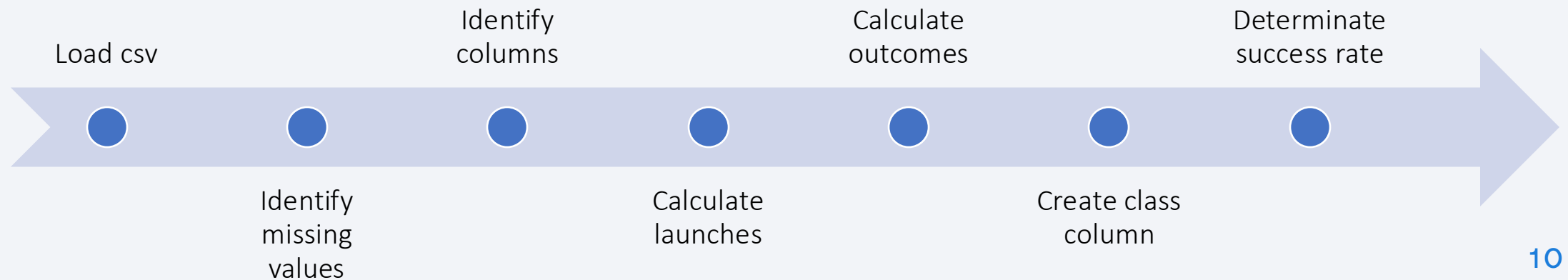
Data Collection - Scraping

- SpaceX Wikipedia page
- Extract tables for Falcon 9 launches
- [GitHub Link](#)



Data Wrangling

- Data load from csv file.
- Identify missing values, columns numerical or categorical.
- Calculate number of launches on sites and orbits.
- Calculate number of mission outcomes of the orbits.
- Create a class column with binary classification from outcome column.
- Determine success rate.
- [GitHub Link](#)



EDA with Data Visualization

- Scatter plot to see the relationship between Flig Number and Payload Mass
- Scatter plot to see the relationship between Flight Number and Launch Site
- Scatter plot to see the relationship between Payload Mass and Launch Site
- Bar plot to see the relationship between success rate and Orbit type
- Scatter plot to see the relationship between Flight Number and Orbit type
- Scatter plot to see the relationship between Payload Mass and Orbit type
- Line plot to see the launch success yearly trend
- [GitHub Link](#)

EDA with SQL

- Performed SQL queries:

- Names of unique launch sites
- 5 records where launch sites begin with 'CCA'
- Total payload mass carried by boosters launched by NASA
- Average payload mass carried by booster version F9 v1.1
- List the date when the first succesful landing outcome in ground pad was achieved
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- List the total number of successful and failure mission outcomes
- List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function
- List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

- [GitHub Link](#)

Build an Interactive Map with Folium

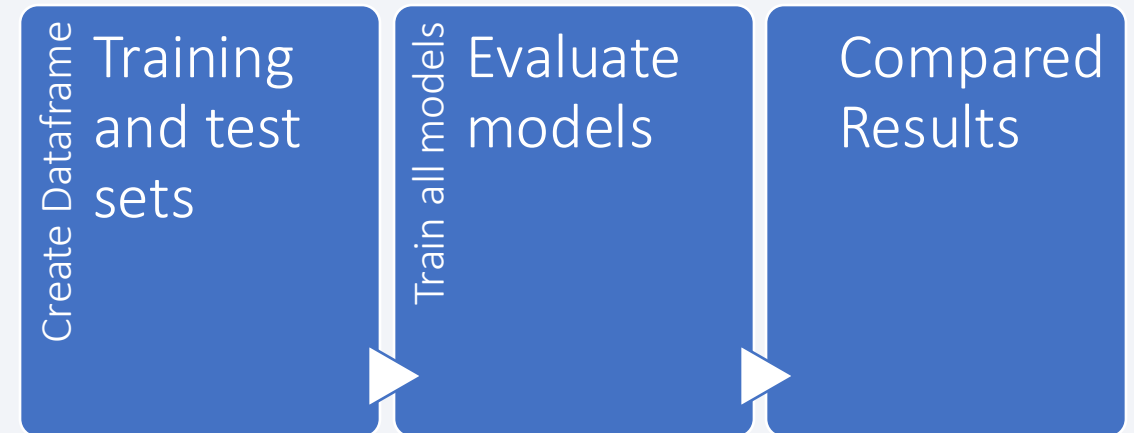
- Map objects created and added to a folium map:
 - Mark all launch sites
 - Circle all launch sites
 - Mark success/failed launches
 - Lines to show distances between two coordinates
- [GitHub Link](#)

Build a Dashboard with Plotly Dash

- Dropdown to select launch sites
- Pie chart for successful launches:
 - All sites show successful launches for sites
 - Site selected show successful/failed launches for this site
- Slider to select Payload Mass range
- Scatter chart to show the correlation between payload and launch success
- [GitHub Link](#)

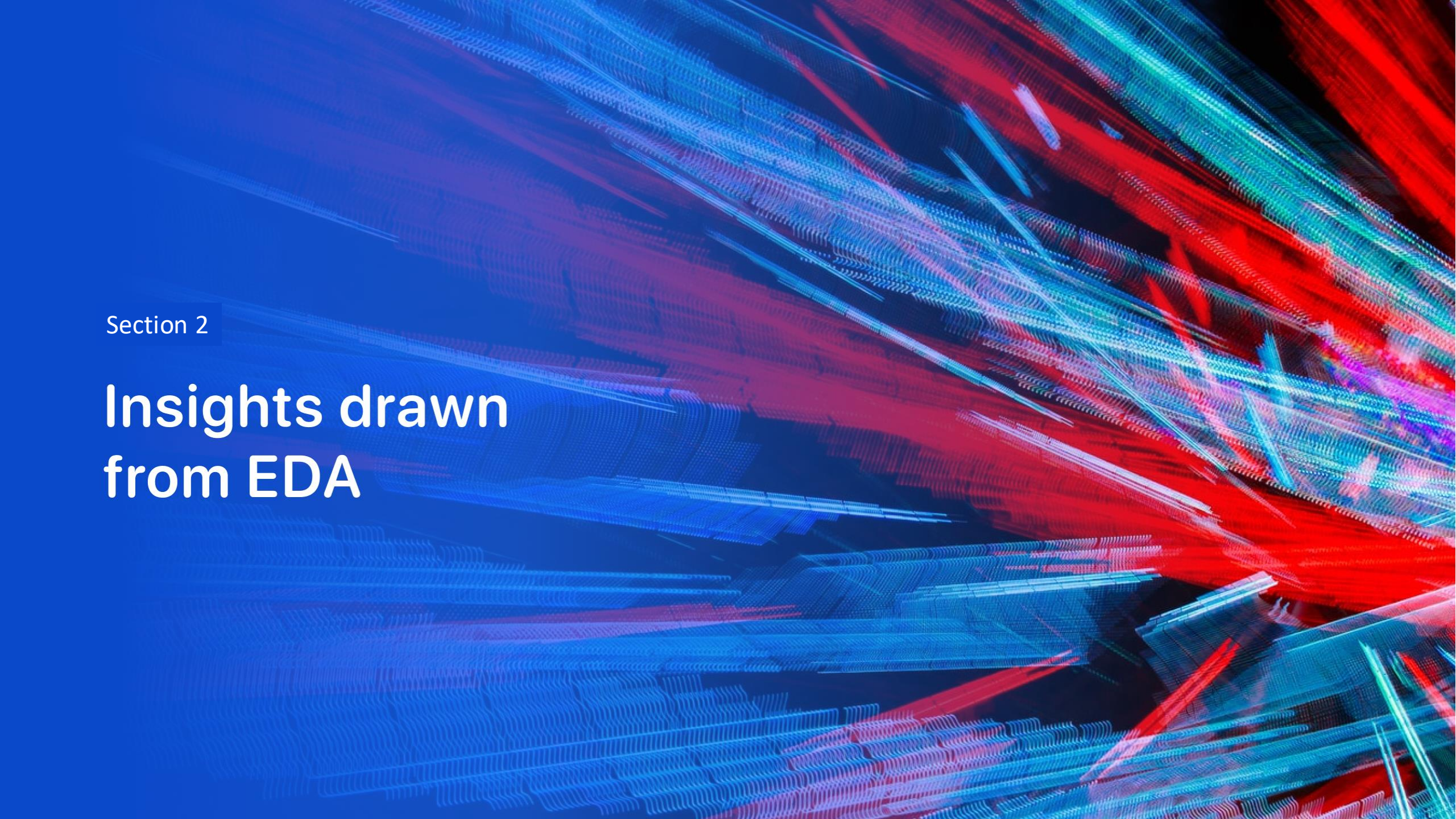
Predictive Analysis (Classification)

- Split data into training and testing sets
- Machine learning models used:
 - Logistic Regression
 - Support Vector Machine
 - Decision Tree
 - K-Nearest Neighbors
- [GitHub Link](#)



Results

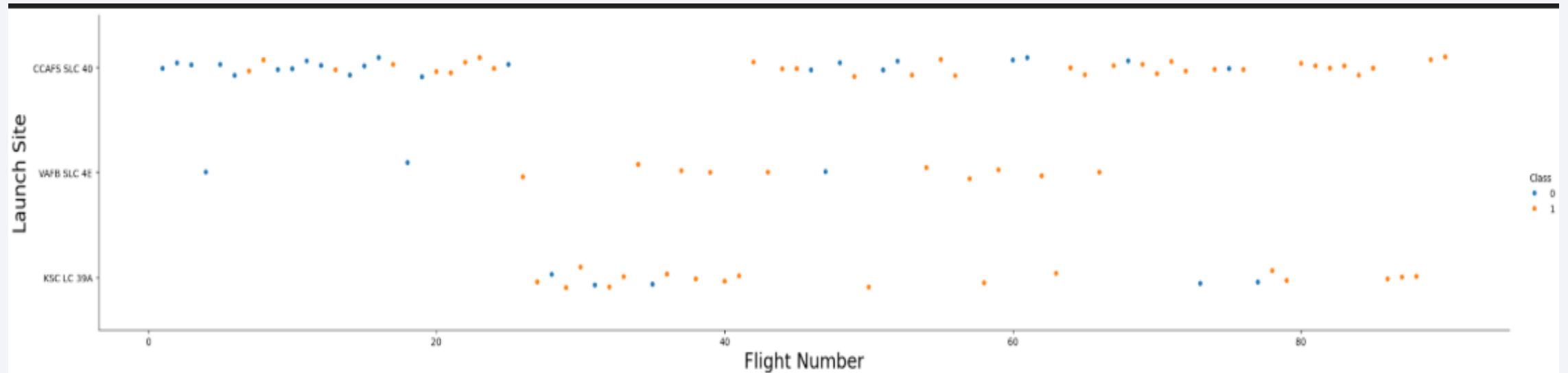
- Exploratory data analysis results
 - Space X uses 4 different launch sites
 - The first success landing outcome happened in 2015 fiver year after the first launch
 - Two booster versions failed at landing in drone ships in 2015: F9 v1.1 B1012 and F9 v1.1 B1015
 - The number of landing outcomes became as better as years passed.

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is dynamic and technological.

Section 2

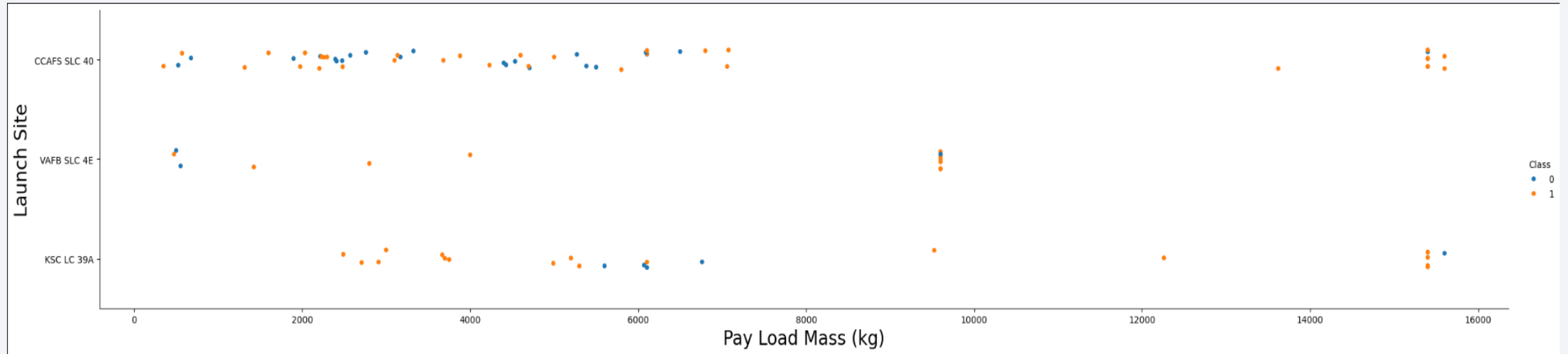
Insights drawn from EDA

Flight Number vs. Launch Site



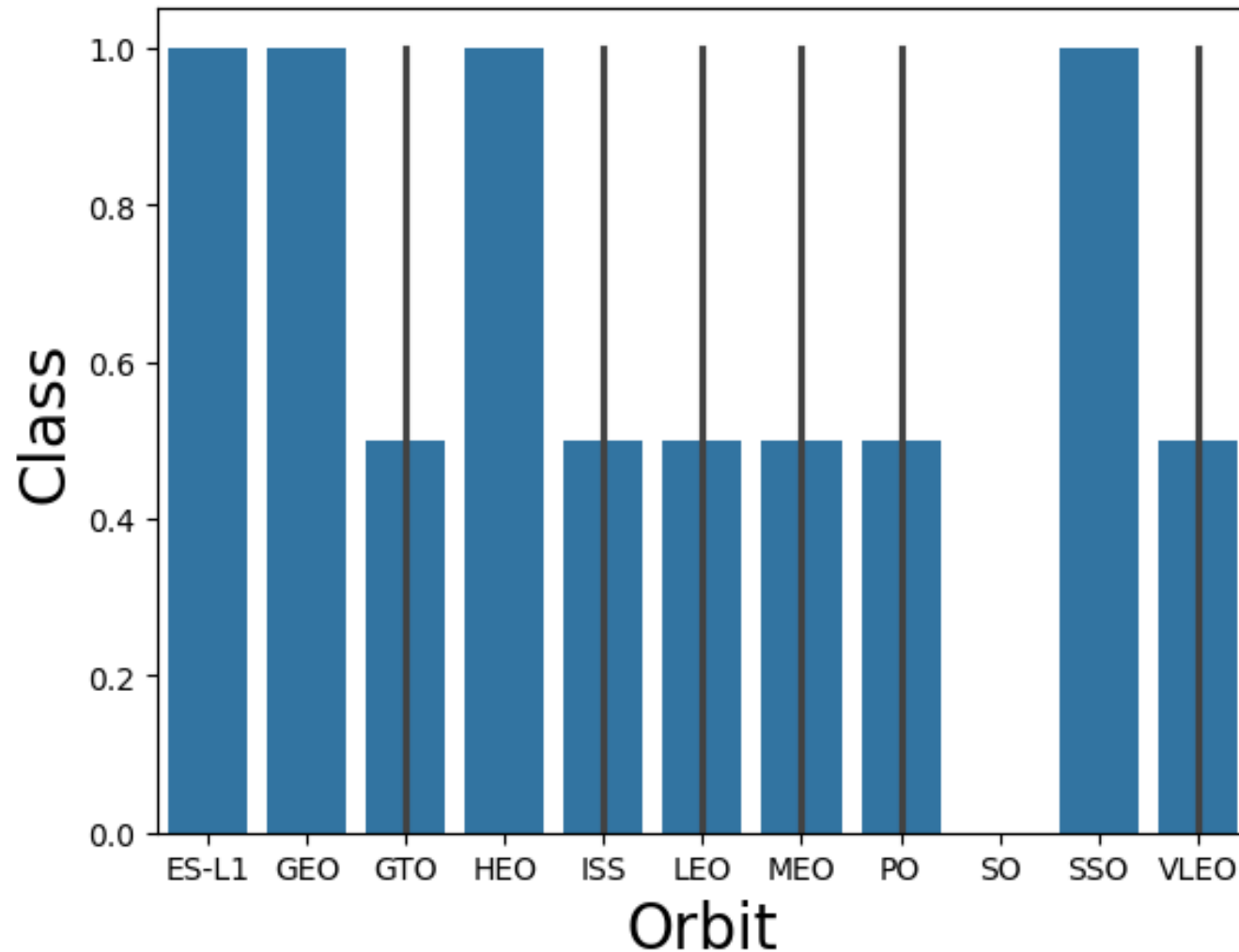
- At more launches better performance

Payload vs. Launch Site



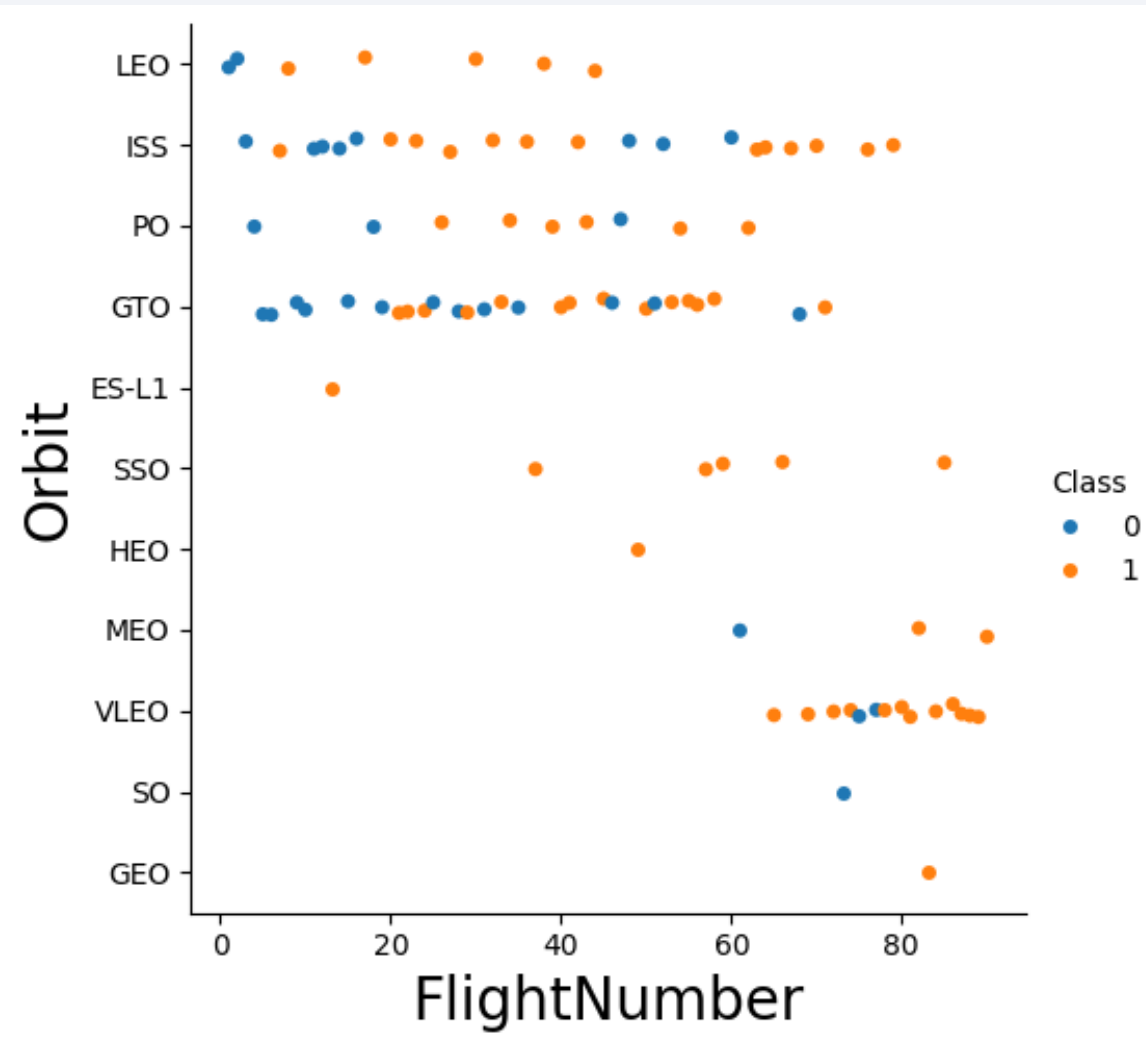
- On increase Payload Mass better performance

Success Rate vs. Orbit Type



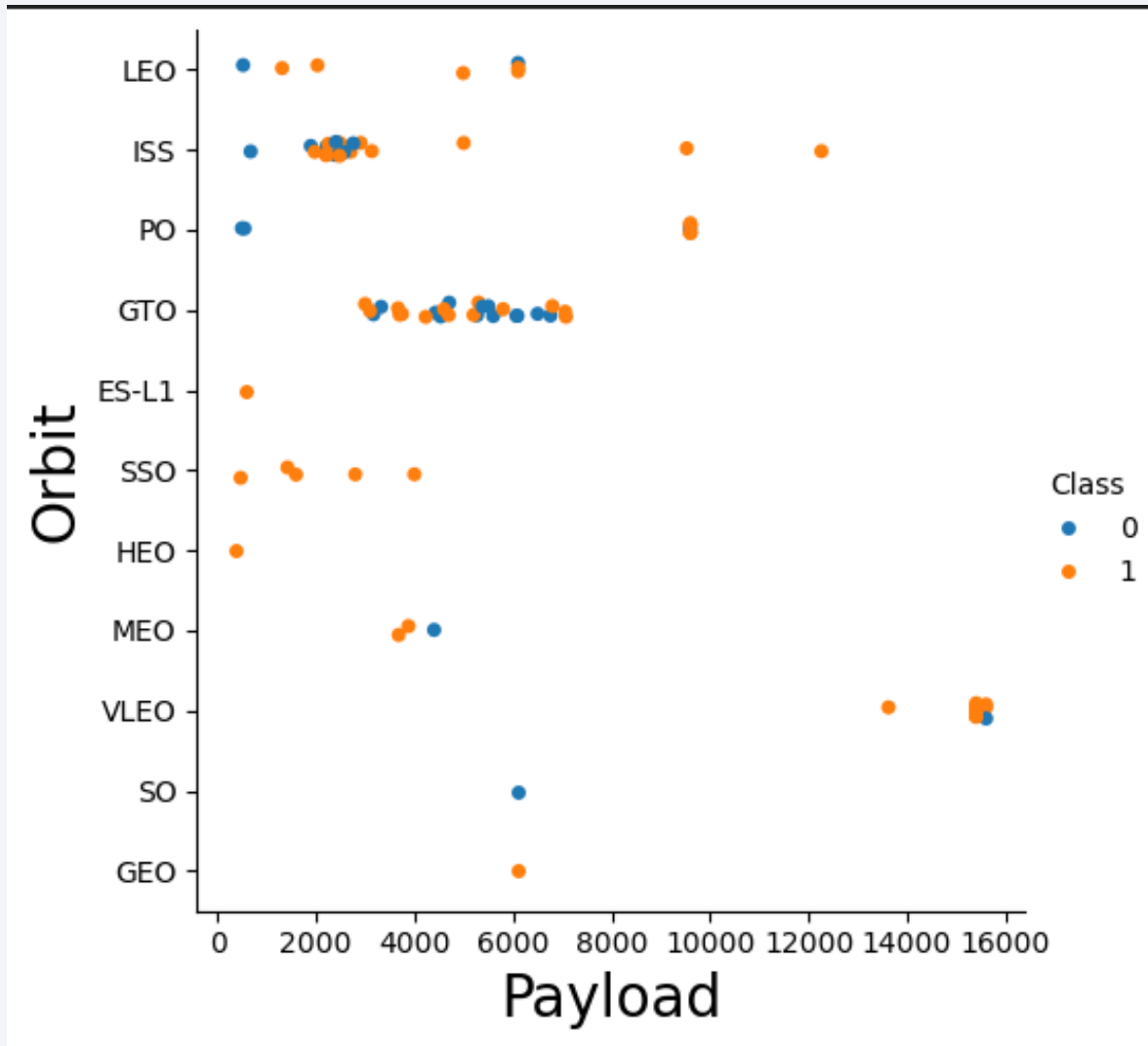
- ES-L1, GEO, HEO and SSO orbits have the highest success rates
- SO have the worst success rate

Flight Number vs. Orbit Type



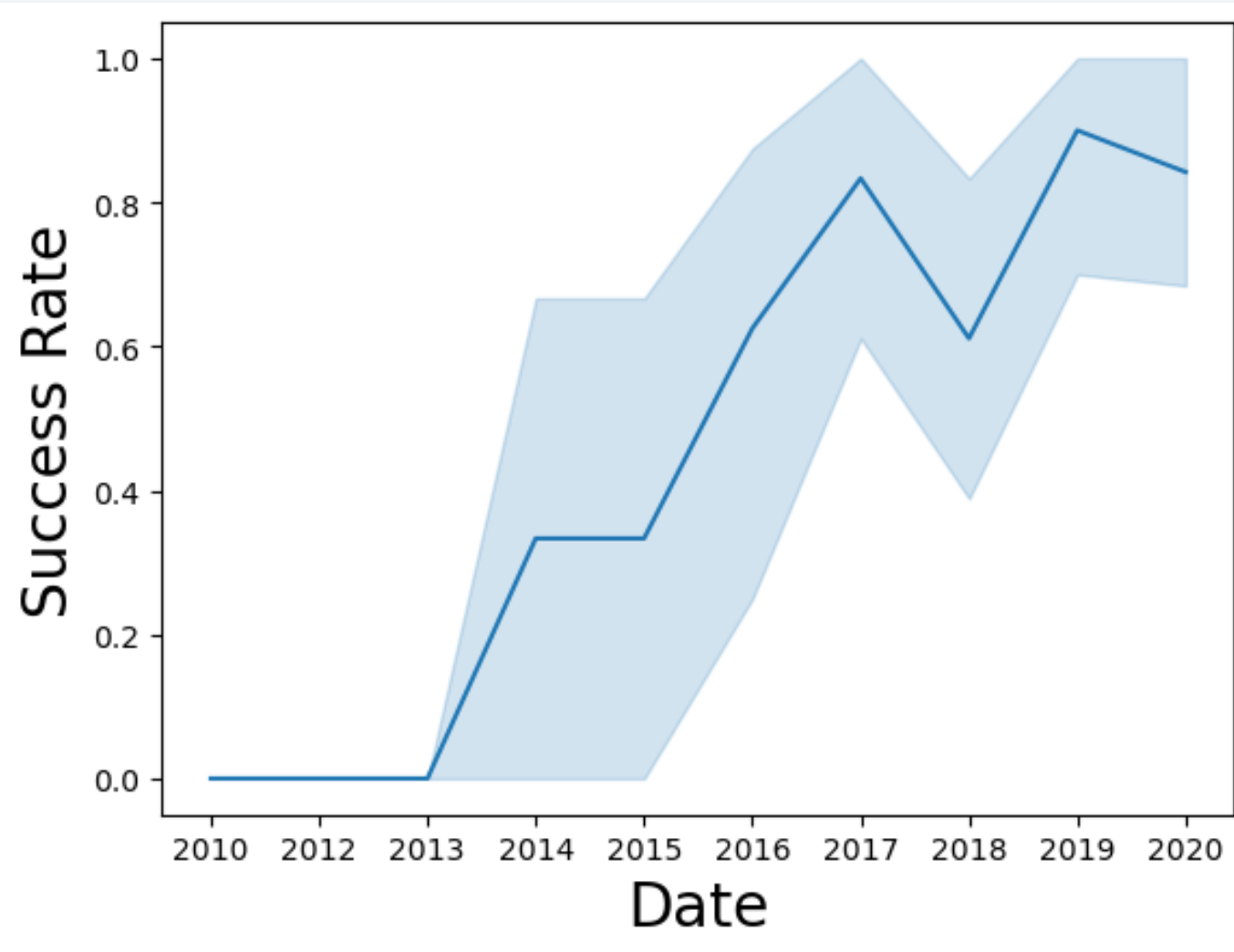
- AT more flight number better performance.
- SSO has the best success rate
- GTO has the worst success rate

Payload vs. Orbit Type



- LEO, ISS and PO with heavy payloads have good landing rates
- The rest of orbits not have an obvious relationship with payloads

Launch Success Yearly Trend



- The success rate have increase from 2013 to 2020
- 2017, 2019 and early 2020 are bad years

All Launch Site Names

- The four launch sites:

```
%sql select distinct Launch_Site from SPACEXTABL;  
✓ 0.0s  
* sqlite:///my\_data1.db  
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with 'CCA'

```
%sql select * from SPACEXTABL where Launch_Site like 'CCA%' limit 5; "SPACEXTABL": Unknown word.
```

✓ 0.0s Python

* [sqlite:///my_data1.db](#)

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Calculate the total payload carried by boosters from NASA

```
%sql select sum(PAYLOAD_MASS_KG_) AS 'NASA Total Payload Mass Carried' from SPACESTABL where Customer = 'NASA (CRS)';
✓ 0.0s
* sqlite:///my_data1.db
Done.
```

NASA Total Payload Mass Carried
45596

- The result is 45596 kg

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1

```
%sql select avg(PAYLOAD_MASS_KG_) AS 'F9 v1.1 Average Payload Mass Carried' from SPACEXTABL where Booster_Version = 'F9 v1.1';
✓ 0.0s
* sqlite:///my_data1.db
Done.
```

F9 v1.1 Average Payload Mass Carried
2928.4

- The average payload mass carried by F9 v.1.1 are 2928.4 kg

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad

```
%sql select min(Date) AS 'Date' from SPACEXTABL where Landing_Outcome = 'Success (ground pad)'
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

Date
2015-12-22

- The first successful landing outcome on ground pad was December 22, 2015

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
%sql select Booster_Version AS 'Name' from SPACEXTABL where Landing_Outcome = 'Success (drone ship)' and PAYLOAD_MASS_KG_ between 4000 and 6000;
✓ 0.0s
* sqlite:///my_data1.db
Done.
```

Name
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes

```
%sql SELECT (SELECT count(*) FROM SPACEXTABL WHERE Mission_Outcome LIKE '%success%') AS "Success", count(*) AS "Failure" FROM 'SPACEXTBL' WHERE Mission_Outcome NOT LIKE '%success%';
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

Success	Failure
100	1

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass

```
%sql select Booster_Version AS 'Name', PAYLOAD_MASS_KG_ AS 'Payload Mass (kg)' from SPACEXTABLE where PAYLOAD_MASS_KG_ = (select max(PAYLOAD_MASS_KG_) from SPACEXTABLE)
✓ 0.0s

* sqlite:///my_data1.db
Done.
```

Name	Payload Mass (kg)
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

```
%sql select substr(Date, 6, 2) AS 'Month', Booster_Version AS 'Name', Launch_Site AS 'Launch Site', Landing_Outcome AS 'Landing Outcome' from SPACEXTABLE where substr(Date, 1, 4) = '2015' and Landing_Outcome = 'Failure (drone ship)'
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

Month	Name	Launch Site	Landing Outcome
01	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

```
%sql SELECT Landing_Outcome AS 'Landing Outcome', COUNT(*) AS Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Count Desc
✓ 0.0s
* sqlite:///my\_data1.db
Done.
```

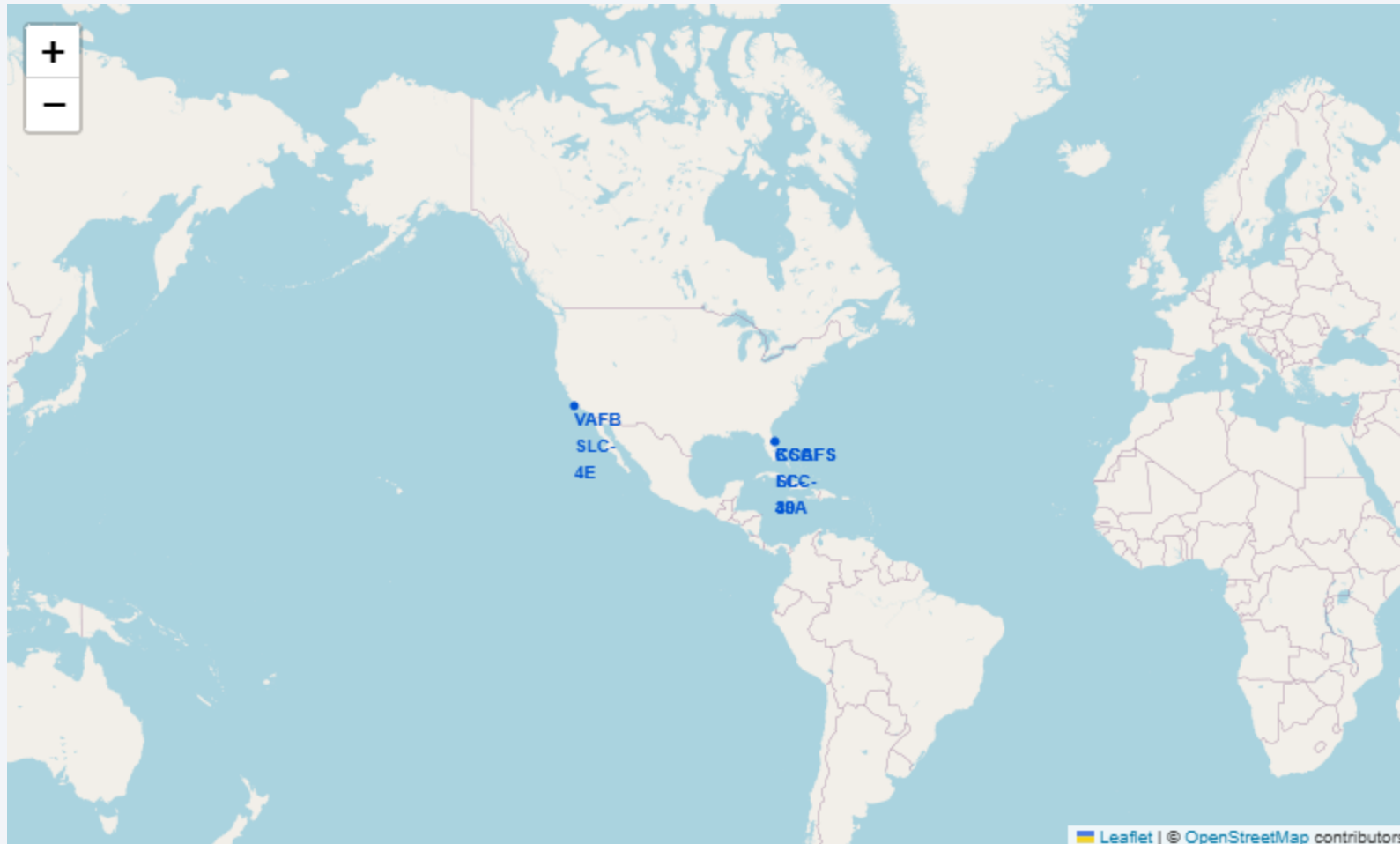
Landing Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark blue, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a curved line separating the dark surface from the blackness of space.

Section 3

Launch Sites Proximities Analysis

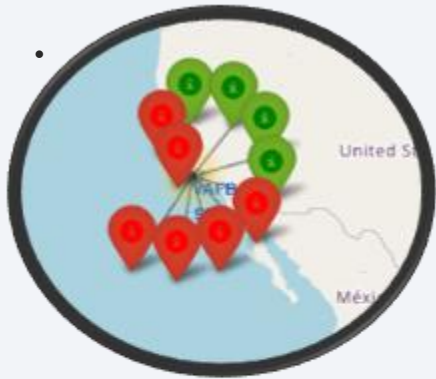
All launch sites



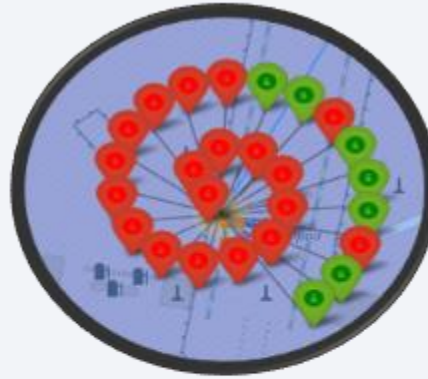
- Launch sites are near to the coast
- Launch sites are near equator line
- Launch sites are far of cities, highways and railways

Launch Outcomes

- Explore the folium map and make a proper screenshot to show the color-labeled launch outcomes on the map



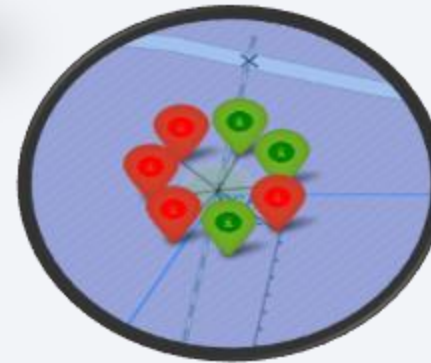
VAFB SLC-4E



CCAFS LC-40



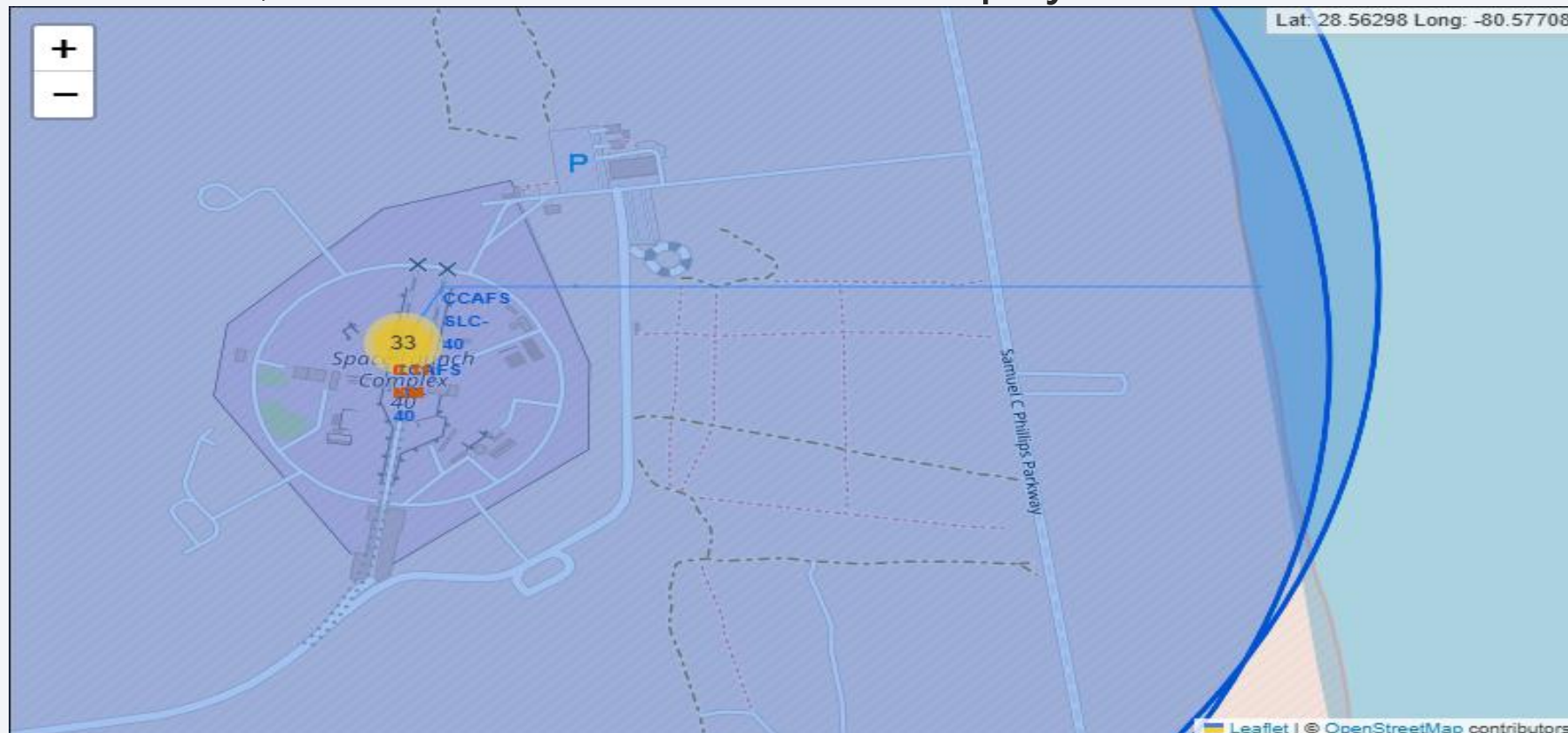
KSC LC-39A

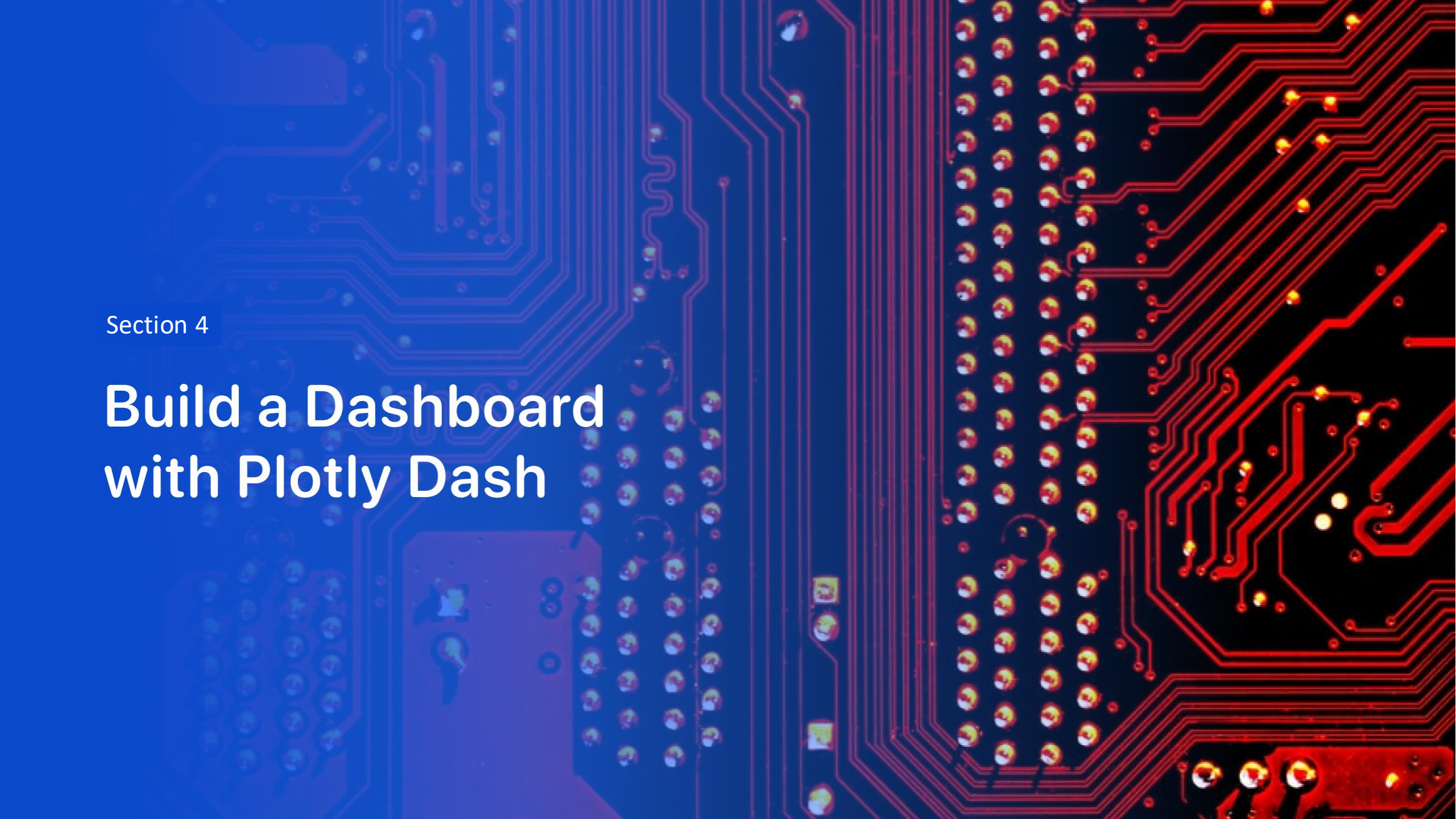


CCAFS SLC-40

Launch site distance

- Explore the generated folium map and show the screenshot of a selected launch site to its proximities such as railway, highway, coastline, with distance calculated and displayed

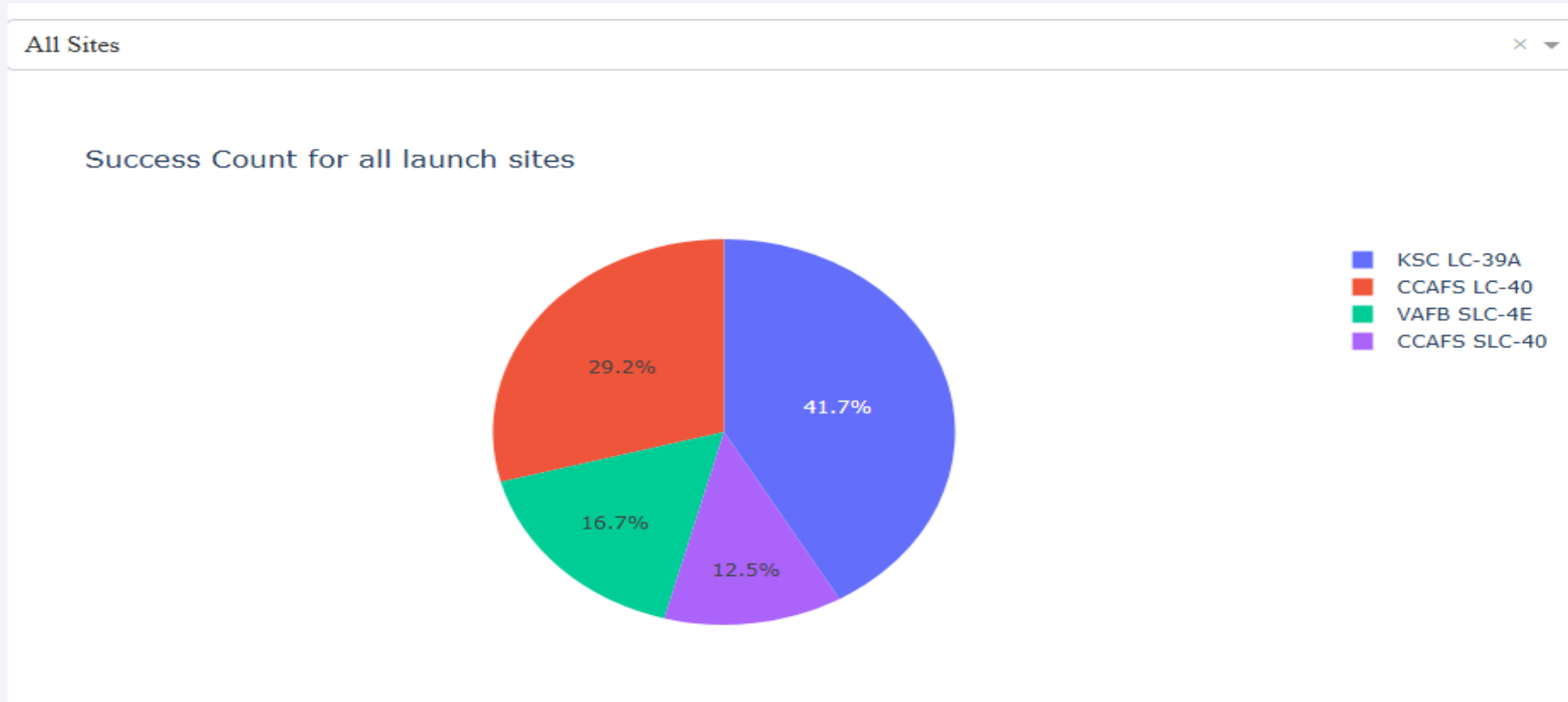




Section 4

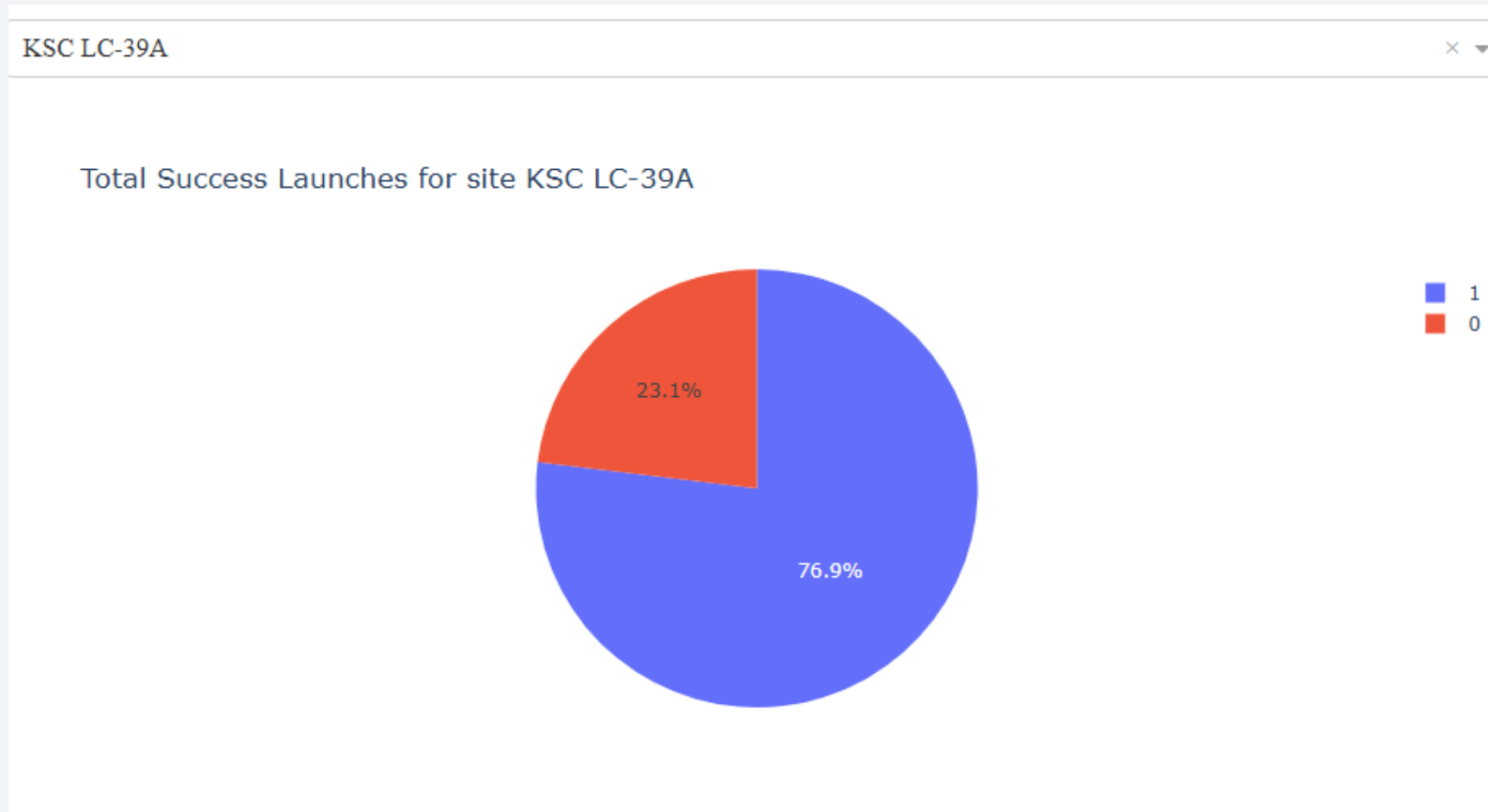
Build a Dashboard with Plotly Dash

Launch Success Count for All Sites



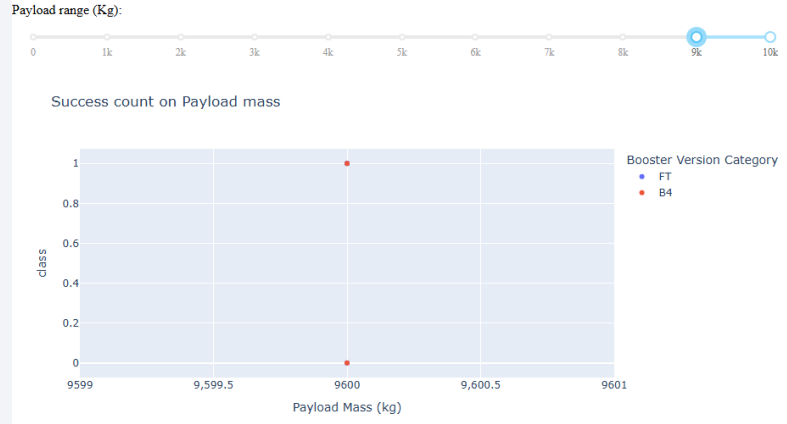
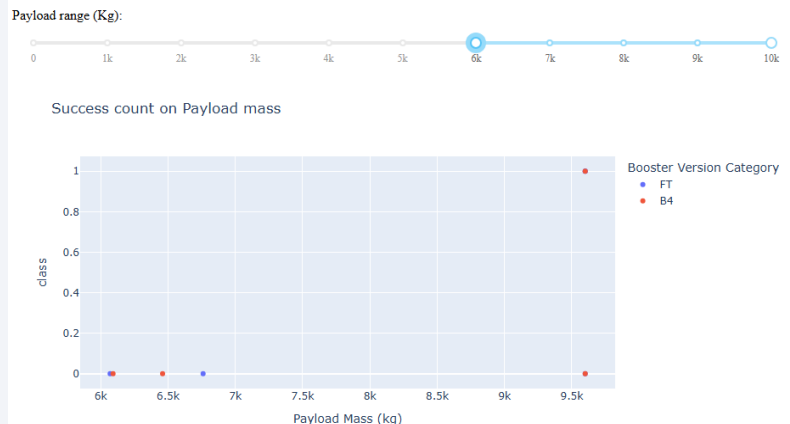
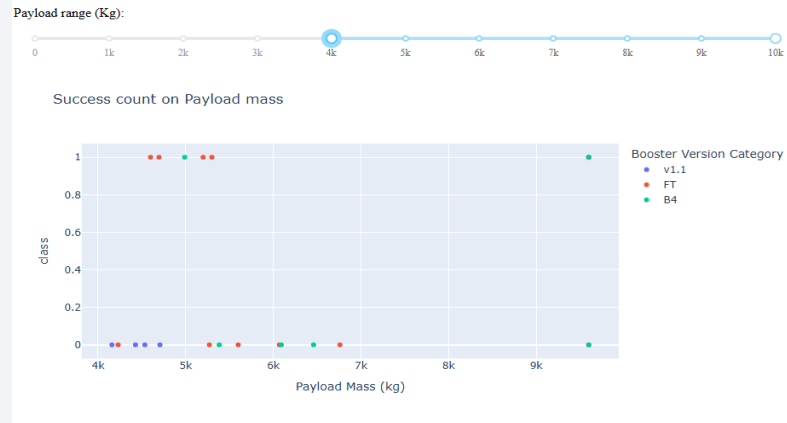
- KSC LC-39A has best success rate

Site with Highest Launch Success Ratio



- With 76.9 KSC LC-39A has best success rate

Payload vs Launch Outcome



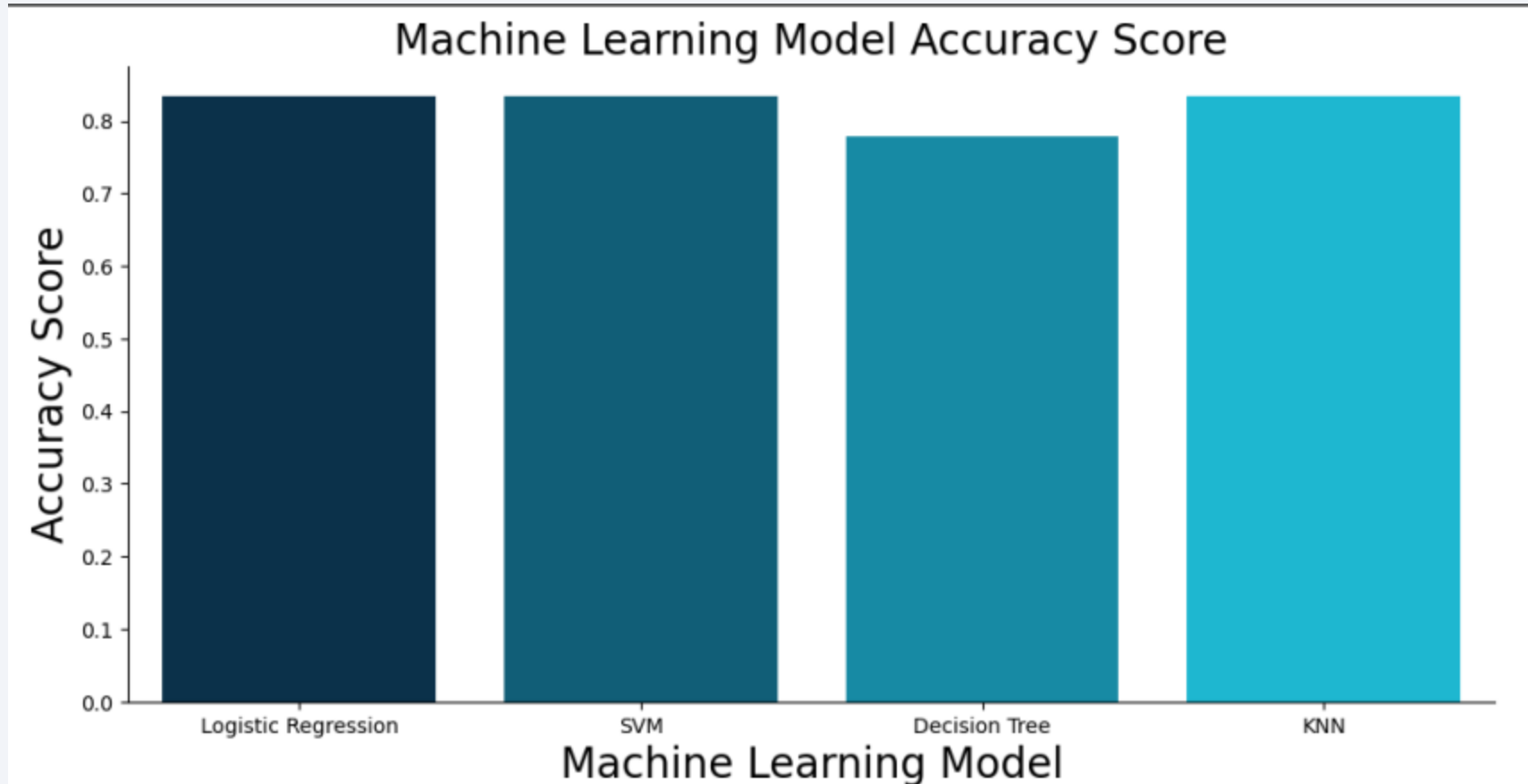
- Range from 2000 to 5000 has best success rate
- After 5000 the success rate go down
- FT booster has best success rate



Section 5

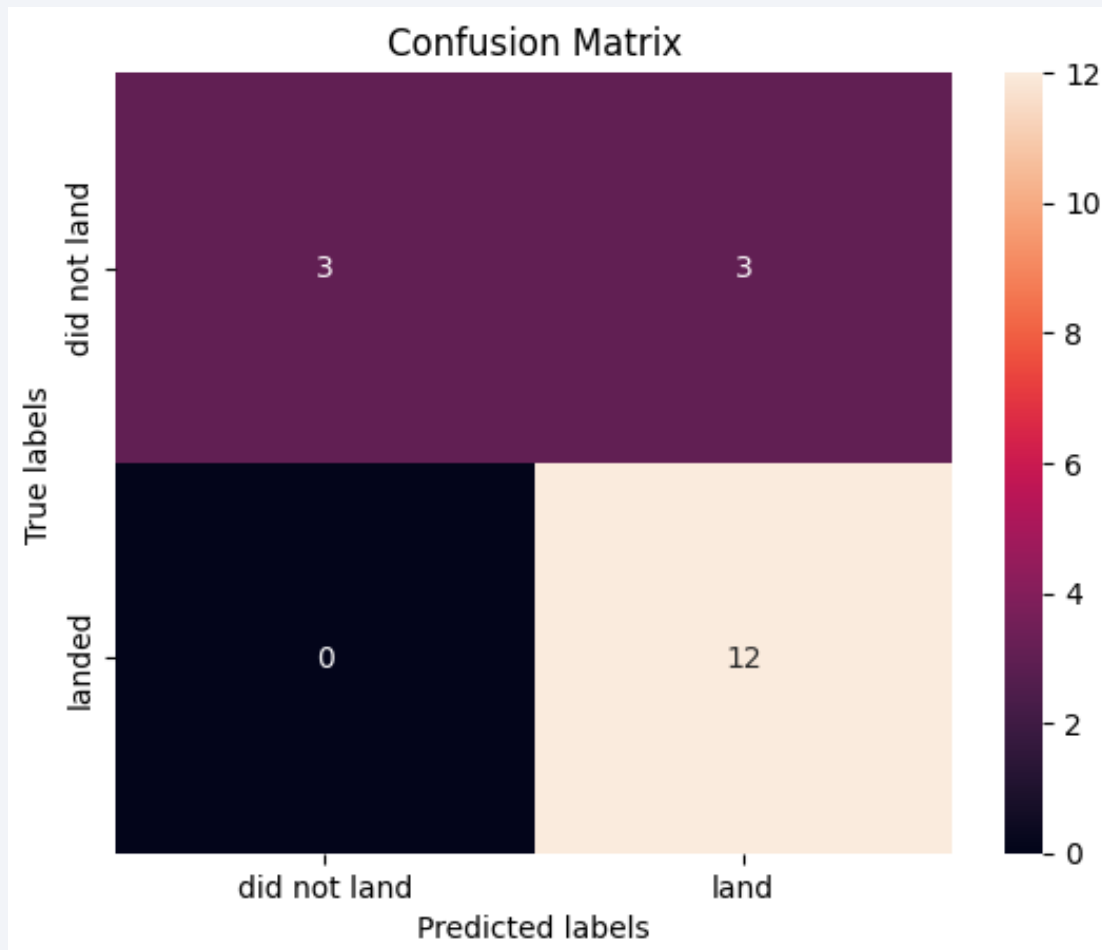
Predictive Analysis (Classification)

Classification Accuracy



- There are no significant differences for all models

Confusion Matrix



- Logistic regression, SVM and KNN have same performance.

Conclusions

- Success rate improves over time among all launch sites
- KSC LC-39A is the best launch site among four launch sites
- Launch success rate started to increase in 2013 till 2020
- Launch Sites are close to sea, not over populated
- Orbits ES-L1, GEO, HEO, SSO, VLEO had the most success rate

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

